

Modulation Classification for Non-orthogonal Multiple Access System using a Modified Residual-CNN

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Abstract—Non-orthogonal multiple access (NOMA) is a promising solution to the problem of spectrum deficiency. NOMA receivers require information about the modulation type of the co-scheduled user's signal to perform successive interference cancellation (SIC). Automatic modulation classification (AMC) is used to reduce the signal overhead created due to sharing of this information. In this paper, a convolutional neural network (CNN) with a modified residual block (MR-CNN) is proposed for AMC in NOMA systems. The classification performance of MR-CNN is evaluated on the input signal with four different modulation formats at varying signal to noise ratios (SNRs). Experimental results demonstrate the proposed model can achieve more than 90% classification accuracy for SNRs higher than 10dB. Further, comparative analysis depicts that the proposed model outperforms existing methods in terms of classification accuracy at low SNRs and has similar performance at high SNRs.

Index Terms—Non-orthogonal multiple access (NOMA), automatic modulation classification (AMC), convolution neural network (CNN), convolutional residual block.

I. INTRODUCTION

THE next-generation wireless network demands low latency, improved fairness, high reliability, massive connectivity, and high spectrum efficiency. Non-orthogonal multiple access (NOMA) is a new emerging technology that helps to meet these demands [1]. In NOMA, multiple users' signals are superposed in time, frequency, and power domain. As multiple users can use the same frequency simultaneously, it boosts spectrum efficiency promisingly. The signals are superposed with power difference based on the channel conditions of receivers. Higher power is allocated to the user with lower channel gain i.e. far user. To recover the signal, the far user treats the near user's signal as noise. The near user needs to apply successive interference cancellation (SIC) to cancel the far user's signal and recover its signal. To perform SIC, the far user's signal needs to be regenerated, for which the

modulation information of the signal should be known to the near user. This information can be sent to the near user by signaling, which will create extra overhead to the spectrum. Automatic modulation classification (AMC) is a promising solution to reduce the overhead and hence is getting the significant attention of the researchers. The majority of the AMC methods in the literature consider a single user in orthogonal multiple access (OMA) scheme without co-channel interference. This type of AMC methods are surveyed in [2] and [3]. The AMC methods can be broadly classified into likelihood-based (LB) and feature-based (FB) methods. The LB methods use the probabilistic approach for classification, whereas the FB methods employ feature extraction to solve the classification problem. A lot of work has been done on LB methods. In [4], the classification of binary phase shift keying (BPSK) and quadrature PSK (QPSK) is performed using a quasi-hybrid-likelihood-ratio-test (QHLRT) algorithm. To gain the performance improvement over [4], the distance between the log-likelihood of the received data and the expected log-likelihood is minimized in [5]. Authors in [6] proposed a modulation classification algorithm for multiple antenna systems by optimally combining the log-likelihood functions. An HLRT based AMC classifier is proposed in [7] where efficient implementation of expectation maximization (EM) algorithm is done to estimate the channel coefficient and noise variance. Authors in [8] used a squared iterative method with parameter checking to accelerate the convergence rate of the EM algorithm. The LB algorithms perform well for less number of modulations (classes) and high SNRs, but for more number of modulations the performance degrades significantly. Also, these methods are computationally complex and hard to implement practically.

On the other hand, the FB methods have lower computational complexity than the LB methods. For classification of continuous phase modulation signals in [9], features are extracted through cyclic spectrum analysis and are fed to a hybrid classifier based on independent component analysis

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and support vector machine (SVM). In [10], the authors first performed the channel estimation and used this information for modulation classification using higher-order cumulants. Authors in [11], used machine learning (ML) classifiers such as Naive Bayes, Linear Discriminant Analysis (LDA), k-nearest neighbors (KNN) and SVM to classify the digital modulation schemes after extracting the features with Stockwell transform (S-transform). Higher order cumulants are used to extract features in [12], [13] and then fed the features to the SVM classifier. Authors in [14] considered overlapped sources and used channel estimation to separate them. For AMC, maximum likelihood-based multicumulant classification is conducted for each of the sources. In [15], authors present a novel AMC algorithm based on variational mode decomposition (VMD) in presence of non-Gaussian impulsive noise and additive white Gaussian noise. The linearly and circularly modulation schemes are classified in [16] using cyclic cumulants. Sparse signal decomposition is used for AMC in [17] using cyclostationary feature extraction algorithm. In [18], AMC for linearly modulated signals of orthogonal frequency division multiplexing (OFDM) system by using discrete Fourier transform (DFT) and normalized fourth-order cumulant is proposed. Through the literature survey, it can be inferred that the traditional machine learning (ML) algorithms such as Naive Bayes, random forest, KNN, SVM, etc. are not feasible for AMC since they have lower classification accuracy.

Deep learning (DL) has been used extensively for AMC in OMA due to its outstanding performance in image processing domain. DL based methods can extract features automatically from a raw signal and perform classification more efficiently than the LB or traditional ML methods. A convolutional neural network (CNN) based modulation classifier is introduced in [19], which gives better classification accuracy than the traditional ML classifiers. In [20], authors proposed an long short-term memory (LSTM) based classifier for AMC that overshoots the performance of CNN in [19]. To decrease the training time required for CNN, a two-step training approach and transfer learning are utilized in [21]. Authors in [22] converted time domain signal into spectrogram images using short-time discrete Fourier transform and used a Gaussian noise filter to remove noise. The filtered images are fed as input to a CNN based classifier. In [23], two CNN models (GoogleNet and AlexNet) are used for AMC by considering constellation images as input to the classifier. For the fusion of multimodal information, the waveform-spectrum multimodal fusion method is adopted in [24] to realize AMC based on residual networks. In [25], a CNN-LSTM based hybrid model is proposed to make the classifier learn temporal and spatial features. The methods discussed so far consider single-user based modulation classification of OMA and do not consider the interference condition from the co-scheduled user as in the NOMA; hence these methods are not feasible to use for the classification of NOMA signal. The only existing method which performs AMC using fourth-order cumulants in NOMA system is proposed in [26]. To the best of our knowledge, there

is no work reported in literature that propose deep learning for AMC in NOMA. In this work, we propose a CNN architecture with a modified residual block to classify the modulation type of the far user's signal at the near user's end. To demonstrate its effectiveness, the CB method [26] is used as a benchmark. The rest of the paper is organized as follows. In the next section, The system model for the downlink of the NOMA is discussed. Section III introduces the proposed MR-CNN architecture along with a modified residual block. In section IV, the classification performance of the proposed model is analyzed and comparative analysis with the existing methods is performed. Finally, the paper is concluded in section V.

II. SYSTEM MODEL

In this work, the NOMA scheme in downlink is considered, where multiple user's signals are superposed with a power difference. The computing complexity of the modulation classification algorithm increases exponentially with the increase in the number of co-scheduled users. However, the principle of modulation classification remains the same for any number of co-scheduled user, hence the number of co-scheduled users are limited to two in this work for convenience. Considering two user NOMA system, the received signal at time instant $t = 1, 2, \dots, N$, is given as

$$y(t) = h \left[\sqrt{P_f} x_f(t) + \sqrt{P_n} x_n(t) \right] + w(t), \quad (1)$$

where x_f and x_n are the unit energy signals sent by the base station (BS) for far and near user, respectively, h represents the Rayleigh fading channel gain between the BS and the near user terminal and the $w(t)$ is the additive white Gaussian noise (AWGN) with zero mean and variance σ^2 . P_f and P_n are the power allocated to the far and near user, respectively. Generally, two users with larger channel gain difference are co-scheduled and more power is assigned to the far user, which has lower channel gain, i.e., $P_f > P_n$ with constraint $P_f + P_n = 1$. In this scenario, the far user recovers its signal by treating the near user's signal as noise, but the near user needs to apply successive interference cancellation (SIC), i.e., the far user's signal is regenerated and canceled from the received signal [27]. The SIC can be symbol-level interference cancellation (SLIC) or codeword-level interference cancellation (CWIC). Of these two, the CWIC is more popular and requires the modulation type used for far user's signal. Therefore, the CWIC is considered in this work to perform the intra-cell interference cancellation. The modulation type for both users is chosen from a set of modulation formats $M = \{\text{BPSK}, \text{QPSK}, \text{8PSK}, \text{16QAM}\}$ with equal probability. The signal $y(t)$ has in-phase (I) and quadrature (Q) component that are represented as

$$y^I(t) = \Re[y(t)] \text{ and } y^Q(t) = \Im[y(t)], \quad (2)$$

where $\Re[\cdot]$ and $\Im[\cdot]$ depict the real and imaginary components of a complex number respectively. The received signal can be written in form of $2 \times N$ matrix, such as

$$Y = \begin{bmatrix} y^I(0) & y^I(1) & \dots & y^I(N-1) \\ y^Q(0) & y^Q(1) & \dots & y^Q(N-1) \end{bmatrix} = \begin{bmatrix} Y^I \\ Y^Q \end{bmatrix}, \quad (3)$$

where Y^I and Y^Q are the row vectors of $1 \times N$ dimension.

III. PROPOSED METHOD

In this work, a CNN based architecture with a modified residual block, namely MR-CNN, is proposed for AMC. The proposed architecture is shown in Fig. 1. The architecture contains four convolution layers, two average pooling layers, three dense layers, and two modified residual blocks. The Rectified linear unit (ReLU) activation function is used in each convolution and dense layer except the output dense layer with four nodes, where the softmax activation function is used to convert the logits into probabilities. Before feeding to the first convolution layer, the input is reshaped to $(2 \times N \times 1)$. The number of filters in the convolution layer is reduced by factor two with depth, which helps in reducing the number of trainable parameters and also the lesser number of filters in the depth hold only the essence of the data. The two modified residual blocks used in the architecture have the identical structure as shown in Fig. 2. The block contains two parallel convolution layers with 16 filters each. One of the convolution layers has filters of size (1×8) which is in main path while the other layer in skip path, has (1×1) size filters. The (1×1) size of filters is used only to reduce the number of filters from the previous layer so that it can be concatenated to the output of the convolution layer in parallel with it. This layer acts as a skip connection in the block and helps in avoiding the vanishing gradient problem. Instead of using linear activation function with (1×1) filters (as it is done in the convolutional residual network), ReLU activation function is used so that the block can learn non-linear features with parallel connections. The details of operations taking place in the modified residual block are discussed in next paragraph. The average pooling layer is used after each modified residual block to introduce smoothness in the extracted features. The stride is one for convolutional and average pooling layers. The number of layers, filter size and other hyperparameters used in the architecture are optimized using the Keras-tuner.

Let an input P of shape $(r \times s)$ with K feature maps is applied to the residual block and there are F number of filters of dimension $(1 \times L)$ in the convolution layer of the main path. The output of convolution operation of l^{th} filter and m^{th} feature map of the input P in the main path of the block can be written as

$$Q_1^l(i, j) = ReLU \left[\sum_{k=j-\lfloor \frac{L}{2} \rfloor + 1}^{j+\lfloor \frac{L}{2} \rfloor} P_z^m(i, k) \times H^l(j - \lfloor \frac{L}{2} \rfloor + 2 - k - L) \right], \quad (4)$$

where P_z^m is the zero padded form of m^{th} feature map of P . The number of zeros padded at the left and right side of the feature map of P are $\lfloor \frac{L}{2} \rfloor + 1$ and $\lfloor \frac{L}{2} \rfloor$ respectively. Here, $\lfloor . \rfloor$ is the floor function and $H^l(v)$ is v^{th} element of l^{th} filter with $v = 0, 1, \dots, L - 1$. The variables $i = 0, 1, \dots, r - 1$ and $j = 0, 1, \dots, s - 1$ indicate i^{th} row and j^{th} column of the input or output feature map respectively.

Similarly, the output of the convolution operation in the skip connection path having filters of shape (1×1) is given by

$$Q_2^l(i, j) = ReLU [P^m(i, j) \times H^l]. \quad (5)$$

After carrying out the convolution operations as given in Eq. (4) and (5) for all the filters and input feature maps, say, the final output of the main path is Q_1 and that of the skip path is Q_2 . The Q_1 and Q_2 both will have shape of $(r \times s \times 16)$. The Q_1 and Q_2 are concatenated vertically as

$$Q = Q_1 \oplus Q_2 = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix}, \quad (6)$$

where $\oplus$ indicates vertical concatenation. Hence the resultant output Q of the block will have shape $(r \times s \times 32)$.

IV. RESULTS AND DISCUSSION

The process of the dataset generation is discussed in this section and then the performance evaluation and comparative analysis with the existing methods are presented.

Dataset is generated synthetically using the system model given in II. Firstly, raw data is modulated by randomly selecting the modulation formats from the set M for far and near users, and then the modulated symbols are normalized to unit energy for both users. Eq. (1) is used to combine the far and near user signals and generate signals at the receiver end. The values of power ratio and N are varied for experimentation purpose and hence for each pair of value, one dataset is generated by varying SNR in range $-10dB$ to $20dB$ with a step size of $2dB$. The pair of value of power ratio and N is chosen from the set $\{(1.2, 800), (1.5, 800), (2, 200), (2, 800), (4, 100), (4, 200), (4, 400), (4, 800)\}$. Each dataset has three dimensions, $800000 \times 2 \times N$, where the first dimension is the number of signals and each signal has N time-samples of I and Q components.

The dataset is split into training and testing sets in the ratio of 60 : 40. As there are 800000 signals in the dataset, the model is trained on 480000 signals and tested on 320000 signals. The model is trained over 50 epochs with a batch size of 500 and it is compiled using the Adam optimizer with a learning rate of 0.0001. As AMC is a multiclass classification problem, categorical cross-entropy loss is used.

A. Performance Evaluation

The performance of the proposed architecture is evaluated using classification accuracy. Fig. 3 shows the effect of SNR on classification accuracy for different values of N . The accuracy approaches its ideal value with the improvement in signal quality. For $N = 200$, the signals of higher than $8dB$ SNR can be classified correctly with an accuracy of more than 95%. The accuracy can be further improved by increasing the number of samples N in a signal but at the cost of increase in computational complexity. Fig. 4 shows the effect of the power ratio on the proposed method. With the increase in the power ratio, the classification accuracy increases. It is due to the fact that the higher power factor means a larger difference in the powers of far and near users, and because the far user

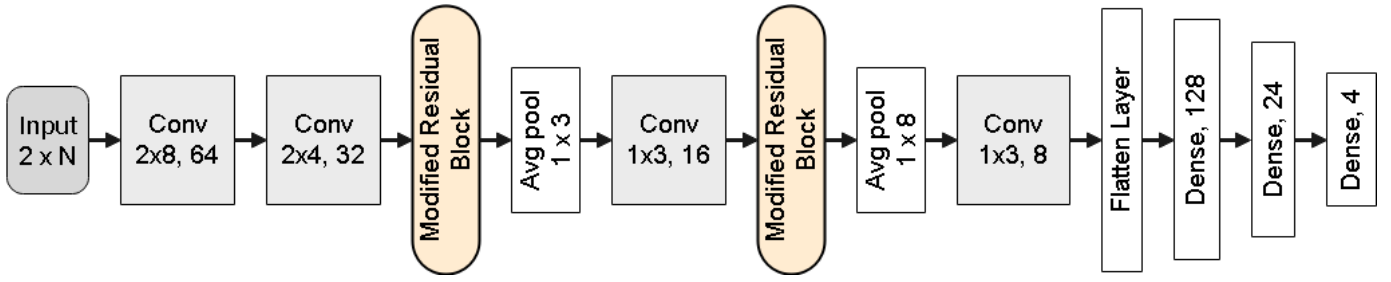


Fig. 1: Proposed Model: MR-CNN.

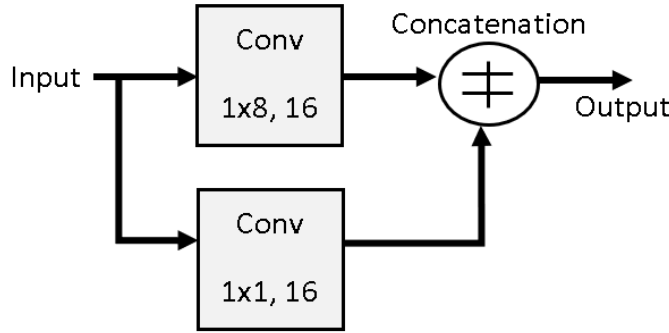


Fig. 2: Modified Residual Block.

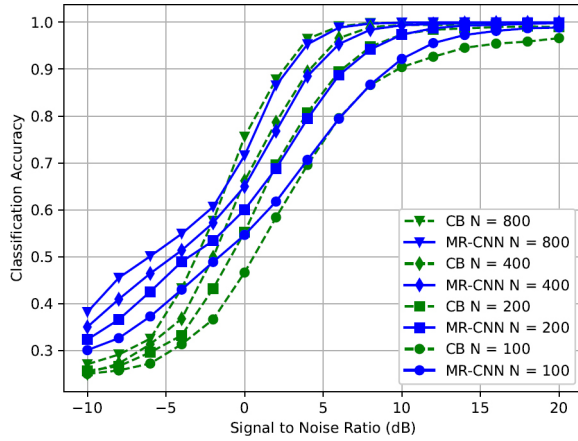


Fig. 3: Classification performance of proposed method and its comparison with cumulant based method for $P_f/P_n = 4$.

has higher power, it will interfere less with the near user's signal. From Fig. 4, it is evident that the proposed method can achieve 100% classification accuracy for a power ratio close to one, although at higher SNR.

Fig. 5 contains three confusion matrices for the proposed model with $N = 200$ and $P_f/P_n = 4$, all are normalized with respect to true labels. The modulation type used for the far user is shown by row labels and the column label shows the predicted modulation type. The elements in the confusion matrix show the probabilities of the predicted modulation

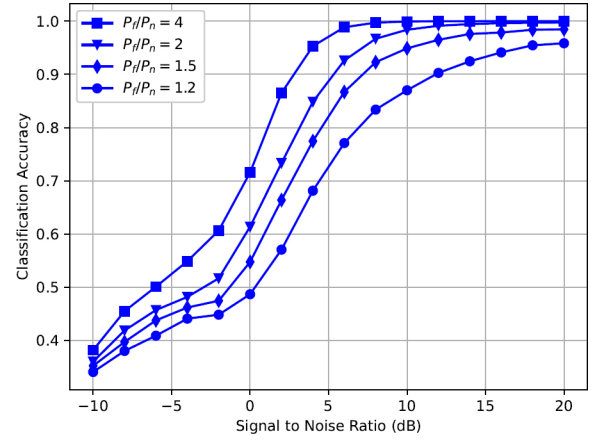


Fig. 4: Classification performance of the proposed method at different power ratios and $N = 800$.

type corresponding to the true modulation type. The diagonal elements show the correct classification probabilities. The confusion matrix in Fig. 5a is for whole test data which includes signals of SNR range -10dB to 20dB with step size of 2dB . If BPSK is employed for the far user at the base station, there will be 91% chance that it will be predicted correctly at the near user end. The classifier could not predict the QPSK, 8PSK, and 16QAM correctly about 30% times, as they have some common constellation points; it is challenging for the model to identify them at low SNRs. The confusion matrix for 0dB SNR is shown in Fig. 5b. The model can accurately predict the BPSK modulation type but the accuracy for other modulations is quite low due to noise. Fig. 5c shows the confusion matrix for signals with 10dB SNR. In this case, the model is able to classify the BPSK and QPSK signals with 100% accuracy. There are only 4% chances that the 8PSK modulation is incorrectly classified as QPSK or 16QAM. Similarly, there are 6% chances that the 16QAM signal is incorrectly classified as QPSK or 8PSK. For higher SNRs, all modulations can be classified with 100% accuracy.

B. Comparative Analysis with [26]

In Fig. 3 and 6, The classification accuracy of the proposed deep learning based method (MR-CNN) is compared with

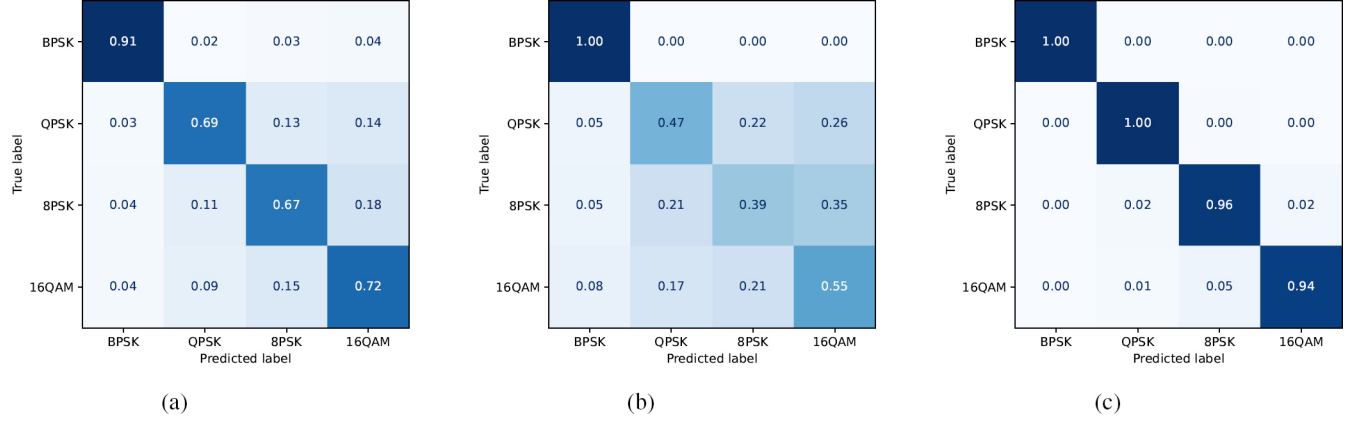


Fig. 5: Confusion matrices for the proposed model with $N = 200$, $P_f/P_n = 4$ and (a) All SNRs, (b) 0 dB SNR and (c) 10 dB SNR.

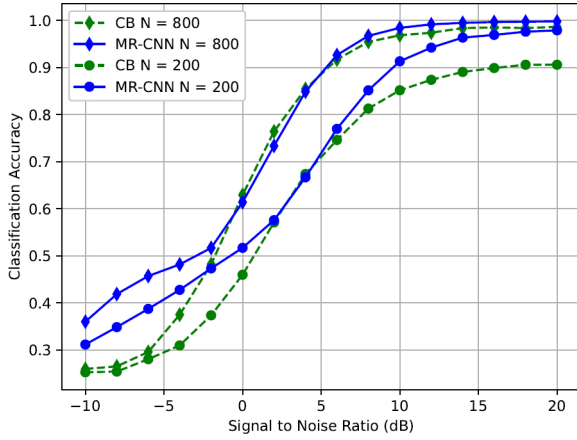


Fig. 6: Classification performance of proposed method and its comparison with cumulant based method for $P_f/P_n = 2$.

the cumulant based (CB) method [26]. The proposed method performs better than the CB method at low SNRs and for SNRs greater than 0 dB, the proposed method achieves almost the same accuracy as of CB method. It is observed that the proposed method has higher classification accuracy than the CB method for lower values of N . Specifically, with $N = 200$ and $P_f/P_n = 2$, The proposed method achieve more than 90% accuracy at 10dB SNR while with CB method 20dB SNR is required to get the same accuracy. If the power ratio is reduced, the difference in the powers of far and near users will be lesser and hence it will be more difficult to distinguish between the modulations. Overall accuracy is the accuracy when test dataset containing all SNRs signals is applied to the classifier. The comparison of the overall accuracy of the cumulant-based (CB) method and the proposed method for different values of P_f/P_n and N is given in Table I. As it is evident from Table I, the proposed model performs better than the cumulant-based method for all pairs of power ratio values and N . The overall

TABLE I: Comparison of Overall Accuracy of the CB Method and the Proposed Method

S.No.	P_f/P_n	N	CB method (%)	Proposed method (%)
1	2	200	62.86	69.19
2	2	800	72.96	76.77
3	4	100	65.76	70.36
4	4	200	71.23	74.96
5	4	400	74.85	78.36
6	4	800	77.93	81.30

accuracy of the proposed method is 6% higher than that of the cumulant-based method with $P_f/P_n = 2$ and $N = 200$. It is observed that the difference between the overall accuracies of the two methods reduces with the increase in N while keeping the power ratio constant.

V. CONCLUSION

In this paper, automatic modulation classification is performed for the far user's signal at the near user's end of the NOMA system. A convolutional neural network with a modified residual block is proposed for classification. The model is trained and tested with a simulated dataset. The classification performance is compared with the only existing method in the literature and it is found that the proposed method is superior to the existing method. The two methods perform almost the same at high SNRs, however, at low SNRs, the proposed method beats the existing method. With the proposed method, 100% classification accuracy can be achieved at high SNR even with a lower number of time-samples of the signal. In the future, we plan to consider the frequency selective fading channel in the simulation of the NOMA downlink and, after including more modulation types, perform modulation classification using an improved deep learning model.

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